UNIT 5

✅ **1 Mark Questions and Answers**

1. **What is a class in Python?**
   → A class is a blueprint for creating objects, bundling attributes (variables) and methods (functions).
2. **What is the purpose of the `__init__()` method in a class?**
   → It is a constructor used to initialize object attributes when an object is created.
3. **What does the `self` keyword represent?**
   → `self` represents the current instance of the class.
4. **Write the syntax to create an object of a class.**

A) obj = ClassName()

5. Differentiate between class attribute and instance attribute.
→ Class attribute is shared by all objects of a class, whereas instance attribute is unique to each object.

6. **Can a class have multiple constructors in Python? (Yes/No)**
→ No, Python does not support multiple constructors directly.

7. Which keyword is used to define a class in Python?
→ `class`

8. What is the default return type of a method if `return` is not used?
→ `None`

Questions from Inheritance:

9. What is inheritance in Python?
→ Inheritance allows a class (child) to acquire properties and methods from another class (parent).

10. Which keyword is used for inheritance in Python?
→ `class Child(Parent):`

11. What type of inheritance allows a class to inherit from more than one parent?
→ Multiple Inheritance.

12. What is method overriding?
→ When a child class defines a method with the same name as the parent class, it overrides the parent's method.

13. Does Python support method overloading? (Yes/No)
→ No, but it can be achieved using default arguments or variable-length arguments.

14. What is the base class of all classes in Python?
→ `object`

`Encapsulation`

1. **What is encapsulation in Python?**
   → Encapsulation is the process of restricting direct access to class variables and methods.
2. **Which symbols are used for private and protected attributes in Python?**
   → `_var` for protected and `__var` for private.
3. **Can private variables be accessed directly outside a class? (Yes/No)**
   → No.
4. **What are getter and setter methods?**
   → Special methods used to access (get) and update (set) private attributes.

`Polymorphism`

5. **What is polymorphism in Python?**
   → The ability of a function/method to take on many forms (same name, different behavior).
6. **Which OOP principle allows method overriding?**
   → Polymorphism.
7. **Give an example of a built-in polymorphic function in Python.**
   → `len()` works for strings, lists, tuples, etc.

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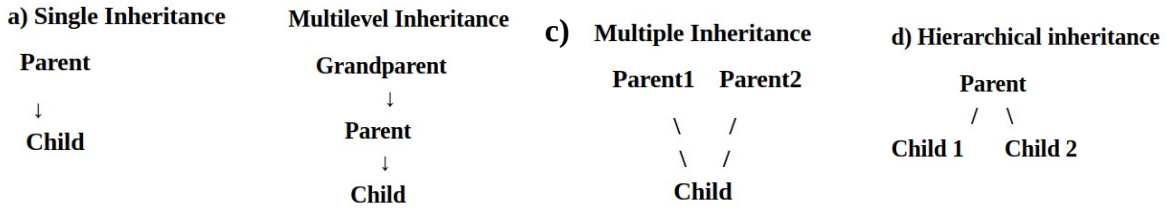
2 Mark Questions and Answers

Types of Inheritance

Type	Description
Single	One parent → One child
Multiple	Many parents → One child
Multilevel	Chain of inheritance
Hierarchical	One parent → Many children



1. **Explain the difference between a class and an object with an example.**
   → A class is a blueprint, and an object is an instance of the class.

```
class Student:   # Student is a class
    def __init__(self, name, age):
        self.name = name
        self.age = age
    def display(self):
        print(self.name, self.age)

s1 = Student("Ram", 19)  # s1 is an object
```

2. Write a simple Python class with attributes.

```
class Student:   # Student is a class
    def __init__(self, name, age): # name and age are attributes
        self.name = name
        self.age = age
    def display(self):
        print(self.name, self.age)
```

3. **How is self used inside a class? Give an example.**
→ It is used to access instance variables and methods.

```
class Student:
    def __init__(self, name):
        self.name = name
```

4. Define a class `Student` with a constructor that initializes `name` and `marks`.

```
class Student:
    def __init__(self, name, marks):
        self.name = name
        self.marks = marks
```

5. Explain the role of methods in classes with an example.
→ Methods define the behavior of objects.

```
class Calculator:
    def add(self, a, b):   # method
        return a + b

c = Calculator()  # Object creation
print(add(4,6))    # calling method
```

Questions from Inheritance:

6. Explain single inheritance with an example.
→ A child inherits from one parent class.

```
class Parent():
    def show_p(self):
        print("I am Parent")

class Child(Parent):
    def show_c(self):
        print("I am Child")

c = Child()  # Object creation with child class
c.show_c()
c.show_p()
```

```
I am Child
I am Parent
```

7. Differentiate between single and multiple inheritance.

- **Single Inheritance** → Child inherits from one parent.
- **Multiple Inheritance** → Child inherits from two or more parents.

8. What is multilevel inheritance? Give an example.
→ In multilevel inheritance, a class is derived from another derived class.

class A: pass
class B(A): pass
class C(B): pass

9. Explain hierarchical inheritance.
→ Multiple child classes inherit from the same parent class.

10. What is the advantage of inheritance?
→ Code reusability and extensibility.

Encapsulation

1. **Differentiate between public, protected, and private members in Python.**
    - Public: Accessible anywhere.
    - Protected (`_var`): Accessible within class and subclasses.
    - Private (`__var`): Accessible only within the class.
2. **Write a simple program to demonstrate private variables.**

class Student:
** def __init__(self, name):**
** self.__name = name # private variable**
s = Student("John")
print(s._Student__name)
Output: John

Polymorphism

3. **Differentiate between method overloading and method overriding.**
    - Overloading: Same method name, different parameters (not directly supported in Python).
    - Overriding: Child class redefines parent class method.
4. **Write an example of polymorphism using a common method.**

```
class Manager:
    def salary(self):
        print("Manager salary: 80000")

class Developer:
    def salary(self):
        print("Developer salary: 60000")

class Intern:
    def salary(self):
        print("Intern stipend: 15000")

# Polymorphism
m = Manager()
m.salary()
d = Developer()
d.salary()
I = Intern()
I.salary()
```

Output:

```
Manager salary: 80000
Developer salary: 60000
Intern stipend: 15000
```

5. Explain the difference between encapsulation and polymorphism.

- **Encapsulation** → Bundles data and methods, restricts access.

  Encapsulation is about *hiding and protecting data*. It means bundling variables (data) and methods (functions) inside a class, and restricting direct access to some of the object's components. Instead of accessing data directly, you use controlled methods (getters/setters).
- **Polymorphism** → Same function name, different behavior.
  Example: `len("abc")` → 3, `len([1,2,3])` → 3.

  Polymorphism is about *one interface, many forms*. It allows the same method or function to behave differently depending on the object or context.

5 Mark Questions and Answers

1. **Explain the concept of class and object in Python with an example program.**

→ A class is a blueprint to create objects; objects are instances of class. Example:

```
class Student:   # Student is a class
    def __init__(self, name, age):
        self.name = name
        self.age = age
    def display(self):
        print(self.name, self.age)

s1 = Student("Ram", 19)  # s1 is an object
s1.display()
```

Output: Ram, 19

2. Write a Python program using a class `Employee` with attributes `name` and `salary`. Include a method to display employee details.

```
class Employee:
    def __init__(self, name, salary):
        self.name = name
        self.salary = salary

    def display(self):
        print(f"Name: {self.name}, Salary: {self.salary}")

e = Employee(“Sam”, 50000)     # Object creation
e.display()                    # Method calling
```

Output: Name: Sam, Salary: 50000

3. Write a program using a class `Calculator` to perform addition, subtraction, multiplication, and division.

```
class Calculator:
    def add(self, a, b): return a + b
    def sub(self, a, b): return a - b
    def mul(self, a, b): return a * b
    def div(self, a, b): return a / b if b != 0 else "Error: Division by zero"

c = Calculator()
print(c.add(10, 5))
print(c.sub(10, 5))
print(c.mul(10, 5))
print(c.div(10, 2))
```

Output:
15
5
50
5

4. Discuss the difference between class variables and instance variables with examples.

- **Class variable** → Shared among all objects.
- **Instance variable** → Unique to each object.

```
class Student:
    college = "MGIT"        # Class variable
    def __init__(self, name):
        self.name = name          # Instance variable

s1 = Student("John")
s2 = Student("Sam")
print(s1.college, s1.name)
print(s2.college , s2.name)
```

Output: MGIT John
MGIT Sam

5. Create a class `Book` with attributes `title`, `author`, and `price`. Write methods to display the book details and check if the book is expensive (price > 500).

```
class Book:
    def __init__(self, title, author, price):
        self.title = title
        self.author = author
        self.price = price

    def display(self):
        print("Title:", self.title)
        print("Author:", self.author)
        print("Price:", self.price)

    def is_expensive(self):
        if self.price > 500:
            print("The book is expensive")
        else:
            print("The book is affordable")

b1 = Book("Python Basics", "Ravi", 650)
b1.display()
b1.is_expensive()
```

Output: Title: Python Basics, Guido, 650
The book is expensive

Questions from Inheritance:

6. **Explain the different types of inheritance in Python with examples.**
   - **Single Inheritance** – One parent, one child.
   - **Multilevel Inheritance –** Inheritance across multiple levels.
   - **Multiple Inheritance** – Child inherits from many parents.
   - **Hierarchical Inheritance** – Many children inherit from one parent.

1. Eg of Single inheritance:

```python
class Person:  # Parent class
    def basic_info(self):
        print("Learning things to find a job in a company")

class Employee(Person):   # Child class
    def role(self):
        print("Attend works assigned by the company")

# Create object of child class
m = Employee()

m.basic_info()     # inherited method
m.role()   # child class method
```

```
Employee Name: Ravi
Role: Manager
```

2. Example of Multilevel Inheritance:

```python
class Person:
    def basic_info(self):
        print("Learning things to find a job in a company")

class Employee(Person):
    def role(self):
        print("Attend works assigned by the company")

class Manager(Employee):
    def decision(self):
        print("Manager makes decisions")

# Create object
s = Manager()

s.basic_info()   # from Person
s.role()       # from Employee
s.decision()     # from Manager or own method
```

```
Learning things to find a job in a company
Attend works assigned by the company
Manager makes decisions
```

3. Example of Multiple Inheritance:

```python
class Father:
    def skill1(self):
        print("Father: Gardening")

class Mother:
    def skill2(self):
        print("Mother: Cooking")

class Child(Father, Mother):   # Inheriting from both
    def skill3(self):
        print("Child: Playing")

# Create object
c = Child()

c.skill1()   # from Father
c.skill2()   # from Mother
c.skill3()   # own method
```

```
Father: Gardening
Mother: Cooking
Child: Playing
```

4. Example of Hierarchical Inheritance:

```python
# Parent class
class Employee:
    def display_company(self):
        print("Company: TCS")

# Child class 1
class Manager(Employee):
    def manager_role(self):
        print("Role: Manager")

# Child class 2
class Developer(Employee):
    def developer_role(self):
        print("Role: Developer")

# Creating objects
m = Manager()
d = Developer()

# Accessing methods
m.display_company()
m.manager_role()

d.display_company()
d.developer_role()
```

```
Company: TCS
Role: Manager
Company: TCS
Role: Developer
```

7. **Explain encapsulation and polymorphism in relation to inheritance.**
   - Encapsulation means **wrapping data (variables) and methods together** and **restricting direct access** to data.
   - In inheritance, **encapsulation controls access** to data.

```python
class Student:
    def __init__(self, marks, grade):
        self._marks = marks  # protected
        self.__grade = grade # private

s = Student(95, 'A+')
print(s._marks)   # to access protected variable
print(s._Student__grade)    # to access private variable
```

```
95
A+
```

   - **Polymorphism** → Same function name behaves differently in different classes.
   - In inheritance, polymorphism is achieved by **overriding methods**.

```python
class Person:  # Parent class
    def display(self):
        print("Learning things to find a job in a company")

class Employee(Person):   # Child class
    def display(self):
        Person.display(self)
#         super().display()
        print("Attend works assigned by the company")

# Create object of child class
m = Employee()

m.display()     # calling overrided method
```

```
Learning things to find a job in a company
Attend works assigned by the company
```

8. Write a program using multilevel inheritance where `Person` → `Employee` → `Manager`.

```python
class Person:
    def basic_info(self):
        print("Learning things to find a job in a company")

class Employee(Person):
    def role(self):
        print("Attend works assigned by the company")

class Manager(Employee):
    def decision(self):
        print("Manager makes decisions")

# Create object
s = Manager()

s.basic_info()   # from Person
s.role()       # from Employee
s.decision()     # from Manager or own method
```

```
Learning things to find a job in a company
Attend works assigned by the company
Manager makes decisions
```

Polymorphism

9. Write a program to demonstrate how python achieves polymorphism by avoiding method overriding.

Sol: **super().method** can be used to access the methods from the parent class in child class. Or parent class method's can be called in child's class **Parent.method(self)**.

```python
class Person:  # Parent class
    def display(self):
        print("Learning things to find a job in a company")

class Employee(Person):   # Child class
    def display(self):
        Person.display(self)
#         super().display()
        print("Attend works assigned by the company")

# Create object of child class
m = Employee()

m.display()     # calling overrided method
```

```
Learning things to find a job in a company
Attend works assigned by the company
```

10. Discuss how Python achieves polymorphism without method overloading. Give an example.

Sol: Polymorphism overcomes: a) Method overriding and b) Method overloading

a) Python achieves polymorphism by using **super()** method to overcome "method overriding".

b) Python achieves polymorphism without **"method overloading"** by using **default arguments or *args**.

```
class Calculator:
    def add(self, *args):
        return sum(args)

m = Calculator()
print(m.add(5))        # 1 argument
print(m.add(5, 10))    # 2 arguments
print(m.add(5, 10, 15))  # 3 arguments
```

Output:
5
15
30

11. What are the types of Constructors? Explain with examples.

A) Parameterized constructor: Constructor with parameters is known as parameterized constructor. The parameterized constructor takes its first argument as a reference to the instance being constructed known as self and the rest of the arguments are provided by the programmer. **Takes parameters to initialize object with specific values.**

```python
class Calculator:   # Class
    def __init__(self, a, b):   # Parametrized constructor
        self.a = a
        self.b = b
    def add(self):   # Methods: add, sub
        return self.a + self.b
    def sub(self):    # Attributes: a, b
        return self.a - self.b

calc = Calculator(10,5)     # Object1 creation with parameters
print(calc.add())   # calling method with no parameters
print(calc.sub())

calc2 = Calculator(100,40)  # Object2 creation
print(calc2.add())
print(calc2.sub())
```

```
15
5
140
60
```

Default constructor: The default constructor is a simple constructor which doesn't accept any arguments. Its definition has only one argument which is (self) reference to the instance being constructed.

- **Does not take any parameters (except `self`).**
- **Initializes object with default values.**

```
Eg of Default Constructor:
```

```python
class Calculator:   # Class
    def __init__(self):   # Default parameters
        self.a = 10
        self.b = 5
    def add(self):   # Methods: add, sub
        return self.a + self.b
    def sub(self):    # Attributes: a, b
        return self.a - self.b

calc = Calculator()     # Object1 creation with default parameters
print(calc.add())
print(calc.sub())
```

```
15
5
```